

ARNAN MUFLIHADI

Public Sector Consultant · GIS & Spatial Analytics · Infrastructure Planning

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PROFILE

Urban planner and public-sector analyst trained at ITB. Most recently at the Ministry of Public Works and Housing (PUPR–BPIW), helped build the analytical framework for a national food-security infrastructure guideline on track for ministerial issuance. Technical work spans spatial modelling — land-cover classification, spatial autocorrelation, GWR, gravity-based transport analysis, and transit accessibility (r5py & GTFS) — backed by Python and SQL across the full data workflow, turning spatial and demographic data into evidence-based analysis.

EXPERIENCE

Urban & Regional Planning Specialist — Ministry of Public Works & Housing (PUPR–BPIW)

Oct 2025 – Mar 2026

Secretariat Agency, Badan Pengembangan Infrastruktur Wilayah · Bandung · Contract

- Developed the analytical framework for a national guideline on food-security data collection and infrastructure needs assessment (regional diagnostics, benchmarking, multi-scenario projections).
- Designed a five-component planning model (regional profiling, infrastructure performance, strategic issue mapping, scenario planning, investment-needs projection), proposed as the basis for the draft RPIW food-security module.
- Piloted in West Java Province; document on track for issuance as a PUPR Ministerial Circular (Surat Edaran).

GIS Analyst — PT Regal Energy Group

Aug 2024 – Jan 2025

Oil & Gas Sector · Jakarta

- Built a centralized geospatial database of 25+ oil & gas well locations from satellite imagery, GPS, and KESDM datasets.
- Designed a custom mobile field-mapping tool that reduced location-verification errors and sped up field surveys.

Program Logistics Coordinator — UNESCO Circular Cities Expedition (Deakin University × SAPPK ITB)

Oct – Nov 2025

NCP Program Circular City · Bandung

- Coordinated logistics, transport, and schedules for 25 participants (5 international, 20 ITB postgraduates) across a multi-site urban expedition.

Project Assistant (Internship) — PT Summarecon Agung Tbk

May – Aug 2019

Property Development · Bandung

- Ran a water-infrastructure feasibility analysis for a residential development to inform the development decision.

SELECTED PROJECTS

Transit Accessibility Scoring — DKI Jakarta [Work in Progress]

2026

- AHP-weighted, 27-category composite accessibility score for 261 kelurahan (2,713 RW), using r5py transit routing on GTFS data; target: Elsevier Urban Transitions Conference 2026 (contributing author).

Sustainable Urban Mobility Plan (SUMP) — Balikpapan City

2025

- Gravity-based OD modelling and study-boundary mapping (land-cover change + multimodal nodes) to prioritize transport infrastructure.

Greater Bandung Drinking-Water Priority Study (ITB thesis)

2023

- Spatial autocorrelation (Moran's I, LISA) and Geographically Weighted Regression (GWR) to rank district-level infrastructure-investment priorities.

Bandung Airbnb Emerging Hotspot Analysis

2024

- Space-time cube + Getis-Ord Gi* analysis in ArcGIS Pro to detect temporal patterns of short-term-rental growth.

Co-Author — “Ngariung, Ngarempug, Ngabangun” (peer-reviewed)

2025

- Spatial analysis and thematic maps for a study on informal-settlement upgrading, applying the 3N participatory method (Ngariung, Ngarempug, Ngabangun).

EDUCATION

B.Sc. Urban & Regional Planning — Institut Teknologi Bandung (ITB)

2023

- Thesis: prioritization of piped drinking-water services across Greater Bandung using spatial autocorrelation (Moran's I, LISA) and GWR.

Data Science Bootcamp — Pacmann

2023 – 2024

- Python, SQL, machine learning, and statistical modelling, with a focus on data visualization and dashboard design.

TECHNICAL SKILLS

Data & Programming: Python (Pandas, GeoPandas, scikit-learn, r5py), SQL (CTEs, window functions), SPSS, Jupyter

GIS & Spatial: ArcGIS Pro, QGIS, PostGIS, Google Earth Engine, land-cover classification (ML), Moran's I / LISA, GWR, hotspot analysis (Gi*), space-time cube, gravity modelling, transit accessibility (r5py/GTFS), GeoDa

Visualization & Reporting: Tableau, Excel (advanced), dashboard design, executive briefings

Strategy & Policy: policy analysis, infrastructure planning, scenario planning, stakeholder coordination, executive report writing

Languages: Bahasa Indonesia (native) · English (professional working proficiency)